

# Mrugank Dake

 [mindemory.io](https://mindemory.io)

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 [github.com/mindemory](https://github.com/mindemory)

 Hanover, NH

## ACADEMIC HISTORY

### Dartmouth College

Postdoctoral Researcher in Psychological and Brain Sciences

Advisor: Dr. John Murray

Hanover, NH

**June 2026 – Present**

### New York University

PhD in Computational Cognitive Neuroscience

Advisor: Dr. Clayton Curtis

Dissertation: The nature and role of distributed representation in Working Memory

MPhil in Experimental Psychology

Coursework: Linear Algebra for Neural & Cognitive Science, fMRI Analysis Methods, Computational Neuroscience

New York, NY

**May 2026**

**May 2025**

### Indian Institute of Science Education & Research (IISER)

BS-MS Dual Degree (Biology & Physics)

Coursework: Nonlinear Dynamics & Chaos, Statistical Thermodynamics, Linear Algebra, Multivariable Calculus

Tirupati, India

**August 2021**

## SPECIALIZED TRAINING

### • Marine Biology Laboratory (MBL)

Methods in Computational Neuroscience (MCN) Course

Woods Hole, MA

**2026**

### • Neuromatch Academy

Deep Learning Course

Virtual

**2022**

## PUBLICATIONS

- **Dake M.**, Dandekar S., Curtis C. E. (2026). Neural synchrony between prefrontal and visual cortex supports visual working memory. *bioRxiv*
- **Dake M.**, Curtis C. E. (2025). Perturbing human V1 degrades the fidelity of visual working memory. *Nature Communications*.
- **Dake M.**, Muni R., Murugesan N., Gokhale K., Rithika S., Kanniyar D., CP Harikrishnan, PS Aravind, Udayraj S., Kanthika L., Kumar H., D'Souza S., Garg E., Paul A., Rajamani N. (*in prep*). More Than a Squeak: Complex Repertoires and Acoustic Adaptation in *Funambulus* Squirrels.

## POSTERS & TALKS

- **Dake M.**, Dandekar S., Curtis C. E. Working memory synchronizes oscillations within and across cortical areas; *Society for Neuroscience (SfN), San Diego CA, USA* **Nov 2025**
- **Dake M.**, Dandekar S., Curtis C. E. Working memory synchronizes oscillations in visual cortex; *Computational Cognitive Neuroscience (CCN), Amsterdam, Netherlands* **May 2025**
- **Dake M.**, Curtis C. E. Perturbing human V1 degrades the fidelity of visual working memory; *In Working Memory Symposium (Virtual)* **July 2024**
- **Dake M.**, Curtis C. E. Do perturbations to visual cortex impact working memory?; *In Society for Neuroscience (SfN), Washington D.C., USA* **Nov 2023**
- Khare S., Chopra S., Grover H., Vohra T., Mahmood U., Jatwani K., Kaushik D., Pandey A., Bhardwaj S., Tanwar A., Singh J., Shekhar C., **Dake M.**, Krishnamoorthy K., Kumar K. Li-Koff: To Detect and Degrade N-Nitrosamines; *Giant Jamboree (iGEM), Boston, USA* **Nov 2020**
- Amar I., Avadhani K., Bajaj M., Balasubramanian D., Bhagat S., Chutani N., **Dake M.**, Khatri U., Krishna N., Jacob M., Mal S., Mohapatra O., Pal A., Saha D., Tripathy B., Mukherjee R., Rao B.J. CoCa coli: probiotic immunotherapy against colon cancer; *Giant Jamboree (iGEM), Boston, USA* **Nov 2019**

## PRE-DOCTORAL RESEARCH EXPERIENCE

- Statistical and Machine learning approaches to Squirrel Acoustics (Masters' Thesis) **Aug 2020 - Aug 2021**  
*Advisor: Dr. Nandani Rajamani, IISER Tirupati, India*
- EEG and MRS Analysis for underpinning neural correlates of attentional blink **May - Dec 2019**  
*Advisor: Dr. Sridharan Devarajan, IISc Bangalore, India*
- CoCa coli: probiotic immunotherapy against colon cancer **Jan 2018 - Nov 2019**  
*Advisors: Dr. Raju Mukherjee; Dr B J Rao, IISER Tirupati, India*
- Isolation and characterization of neurons and glia from goat hippocampal culture **Sep 2018 - May 2019**  
*Advisor: Dr. Vasudharani Devanathan, IISER Tirupati, India*
- Identifying differentially expressing tumor suppressor genes in *Drosophila melanogaster* wing and haltere imaginal discs **May - Aug 2018**  
*Advisor: Prof. L S Shashidara, IISER Pune, India*
- RNA Origami (Square) using in-vitro replication and transcription **Jan - March 2018**  
*Advisor: Dr. Ashwani Sharma, IISER Tirupati, India*
- Head direction cell model for rodents with internal updating mechanism **May - Aug 2017**  
*Advisor: Dr. Collins Assisi, IISER Pune, India*

## PROFESSIONAL & TEACHING EXPERIENCE

|                                                                        |                           |
|------------------------------------------------------------------------|---------------------------|
| <b>NYU Leslie Entrepreneurial Institute</b>                            | New York, NY              |
| Leslie Founder                                                         | <b>Feb 2026 – Present</b> |
| <b>Meta Reality Labs</b>                                               | Burlingame, CA            |
| Product Data Science (PhD) Intern                                      | <b>May – Aug 2025</b>     |
| <b>New York University</b>                                             | New York, NY              |
| <i>Teaching Assistant:</i> Scientific Programming (MA Course)          | <b>Fall 2025</b>          |
| <i>Teaching Assistant:</i> iEEG/EEG/MEG Methods (PhD Course)           | <b>Fall 2024</b>          |
| <i>Teaching Assistant:</i> Introduction to Cognitive Neuroscience (UG) | <b>Fall 2023</b>          |
| <i>Teaching Assistant:</i> Math Tools III: Linear Systems (PhD Course) | <b>Spring 2023</b>        |

## PROFESSIONAL SERVICE

|                                                                                                                                     |                       |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Ad-hoc Reviewer:</b> eLife, ICLR (2026), Cognitive Science Society (2025), Computational Cognitive Neuroscience (CCN 2025, 2026) |                       |
| <b>Website Design Coordinator:</b> Climatedmatch Academy                                                                            | <b>May – Nov 2023</b> |
| <b>Project Mentor:</b> iGEM, Manav Rachna Institute                                                                                 | <b>2020</b>           |
| <b>Student Team Leader:</b> iGEM, IISER Tirupati                                                                                    | <b>2019</b>           |

## AWARDS

|                                                                           |                    |
|---------------------------------------------------------------------------|--------------------|
| Society for Neuroscience Trainee Professional Development Award (\$1,000) | <b>2025</b>        |
| Dean's Award for Research Communication Excellence, NYU (\$2,000)         | <b>2025</b>        |
| GSAS Conference Travel Grant, NYU                                         | <b>2025</b>        |
| MacCracken Doctoral Fellowship, NYU                                       | <b>2021 – 2026</b> |
| Indian Biological Engineering Competition (iBEC) Grant (~\$12,000)        | <b>2020</b>        |
| Indian Biological Engineering Competition (iBEC) Grant (~\$12,000)        | <b>2019</b>        |
| INSPIRE Fellowship, Department of Science and Technology India            | <b>2016 – 2021</b> |
| 5th Place, Indian National Brain Bee (INBB)                               | <b>2015</b>        |

## SKILLS

### Programming

Python, MATLAB, R, SQL,  $\LaTeX$ , CSS, JavaScript, HTML, React, NodeJS, Supabase, Vercel, Antigravity, VSCode

### Neuroscience Tools

EEG, MEG, fMRI, TMS, ECoG, Fieldtrip, MNE-Python, SimNIBS, Psychtoolbox

### Data Science & ML

PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks, Time-series Analysis

### Other Tools & Languages

Git/GitHub, Adobe Illustrator, Adobe Photoshop  
English (Fluent), Hindi, Marathi